

Functional Interface and Lambda Expression

Exercises

Task 1: Create a Custom Functional Interface

steps -

1. Create a functional interface (e.g., Calculator) with a method like `int calculate (int a, int b)`.

Then, write a lambda expression to perform operations such as addition, subtraction, multiplication etc using interface.

Task 2: Creating and Running a Thread with Lambda

Use a lambda expression to provide the implementation of the Runnable interface.

steps -

1. Create a new Java class (e.g., RunnableWithLambda).

2. Inside the main method:

Use a lambda expression to implement the Runnable interface.

Print a message (e.g., "Thread is running using Lambda!") inside the run method.

Create a Thread object using the Runnable implementation.

Start the thread using the `start()` method.

Task 3: Create a Custom Functional Interface with Default and Static Methods.

Steps -

1. Create a functional interface StringProcessor with a abstract method "process".

2. Add a default method to convert the string to uppercase.

3. Add a static method to reverse a string.

4. Use lambda expressions to implement the process method for:

Removing spaces from a string.

Replacing vowels with *.

Task 4: Use Built-in Functional Interfaces

Steps:

1. Use a Predicate to filter a list of integers and print only even numbers.

2. Use a Consumer to print each element of a list.

3. Use a Function to convert a list of strings to their lengths.